Name_____

Date_____

AP Chemistry
Chapters 6 & 7 Test
Electronic Structure & Periodic Properties
Practice Test A

Section I: Multiple Choice, 40 questions, written to be done without a calculator. Choose the best answer and write its letter in the blank.

Section II: Free Response, 2 questions (20 points). Calculator allowed. For full credit, write the formula, substitute with units, and give the answer with units and correct significant figures. Answer "explain" parts in complete sentences.

A periodic table is on the page after the test. Constants are not provided.

Section I: Multiple Choice

Directions: Each question is followed by four answer choices. Select the one that is best in each case.

_____ **1.** Which statement about electromagnetic radiation in a vacuum is correct?

(A) Wavelength and frequency are directly proportional.
(B) Radio waves have a higher frequency than X rays.
(C) All electromagnetic radiation travels at the same speed.
(D) Higher-frequency radiation has a longer wavelength.

_____ **2.** Which list is in order of increasing energy per photon?

(A) radio < microwave < infrared < ultraviolet
(B) ultraviolet < visible < infrared < radio
(C) X ray < ultraviolet < visible < gamma
(D) infrared < radio < visible < ultraviolet

_____ **3.** Light with a wavelength of 6.00×10^{-7} m has what frequency? ($c = 3.00\times10^{8}$ m/s)

(A) 2.00×10^{-15} s^{-1}
(B) 1.80×10^{2} s^{-1}
(C) 5.00×10^{15} s^{-1}
(D) 5.00×10^{14} s^{-1}

_____ **4.** Which expression gives the energy, in joules, of one photon with a wavelength of 400. nm?

(A) $(6.626\times10^{-34})(2.998\times10^{8}) / (400.)$
(B) $(6.626\times10^{-34})(2.998\times10^{8}) / (4.00\times10^{-7})$
(C) $(6.626\times10^{-34})(4.00\times10^{-7}) / (2.998\times10^{8})$
(D) $(2.998\times10^{8}) / [(6.626\times10^{-34})(4.00\times10^{-7})]$

_____ **5.** Photon X has a wavelength of 300 nm and photon Y has a wavelength of 600 nm. Compared with photon Y, photon X has

(A) half the energy and twice the frequency
(B) twice the energy and half the frequency
(C) twice the energy and twice the frequency
(D) the same energy, because both travel at c

_____ **6.** Green light ejects electrons from a certain metal. Which light will also eject electrons from it?

(A) blue light
(B) red light
(C) infrared light
(D) microwave radiation

_____ **7.** Light above the threshold frequency shines on a metal. Making the light brighter (more intense) will

(A) increase the kinetic energy of each ejected electron
(B) raise the threshold frequency
(C) have no effect at all
(D) increase the number of electrons ejected per second

_____ **8.** The idea that energy is absorbed or emitted only in specific packets, each equal to hν, is called

(A) quantization of energy
(B) the uncertainty principle
(C) the Aufbau principle
(D) Hund's rule

_____ **9.** A line spectrum differs from a continuous spectrum because a line spectrum

(A) contains all visible wavelengths
(B) is produced only by heated solids
(C) contains only specific wavelengths that are unique to each element
(D) appears only in the ultraviolet region

_____ **10.** An atom's van der Waals (nonbonding) radius is larger than its bonding radius because

(A) nonbonded atoms only touch, so their electron clouds don't overlap
(B) bonded atoms gain extra protons
(C) bonded atoms lose electrons
(D) van der Waals radii are measured on ions

_____ **11.** In a hydrogen atom, which electron transition emits light of the longest wavelength?

(A) $n = 2 \rightarrow n = 1$
(B) $n = 3 \rightarrow n = 2$
(C) $n = 4 \rightarrow n = 3$
(D) $n = 4 \rightarrow n = 1$

_____ **12.** An electron in a hydrogen atom moves from n = 2 to n = 5. Which describes ΔE and the light?

(A) ΔE is negative; light is emitted
(B) ΔE is positive; light is emitted
(C) ΔE is negative; light is absorbed
(D) ΔE is positive; light is absorbed

_____ **13.** Using $E_n = -2.18\times10^{-18}$ J $(1/n^2)$, the energy of a hydrogen electron in n = 2 is

(A) -1.09×10^{-18} J
(B) -5.45×10^{-19} J
(C) $+5.45\times10^{-19}$ J
(D) -8.72×10^{-18} J

_____ **14.** Which is a limitation of the Bohr model?

(A) It cannot explain why hydrogen has a line spectrum.
(B) It works only for hydrogen and other one-electron species.
(C) It states that energy levels are quantized.
(D) It states that electrons can move between energy levels.

_____ **15.** The uncertainty principle states that

(A) an electron's exact position and exact speed cannot both be known at the same time
(B) no two electrons can have the same four quantum numbers
(C) electrons fill orbitals from lowest to highest energy
(D) energy is emitted when an electron drops to a lower level

_____ **16.** Which set of quantum numbers (n, l, m_l, m_s) is allowed?

(A) (2, 2, 0, +½)
(B) (3, 1, −2, −½)
(C) (1, 0, 0, 0)
(D) (4, 3, −3, +½)

_____ **17.** What is the maximum number of electrons in the n = 4 shell?

(A) 8
(B) 16
(C) 32
(D) 18

_____ **18.** How many orbitals are in a 5d subshell?

(A) 5
(B) 2
(C) 3
(D) 10

_____ **19.** Which quantum number determines the shape of an orbital?

(A) n
(B) m_l
(C) l
(D) m_s

_____ **20.** An electron in a 3p orbital has which values of n and l?

(A) n = 3, l = 0
(B) n = 3, l = 1
(C) n = 3, l = 2
(D) n = 1, l = 3

_____ **21.** In a lithium atom, which orbital is lower in energy, and why?

(A) 2p, because it is farther from the nucleus
(B) They are equal, because both have n = 2
(C) 2p, because p orbitals have more lobes
(D) 2s, because it penetrates closer to the nucleus and feels a larger Zeff

_____ **22.** Which 2p orbital diagram for a nitrogen atom breaks Hund's rule?

(A) [↑] [↑] [↑]
(B) [↓] [↓] [↓]
(C) [↑↓] [↑] [ ]
(D) none of these

_____ **23.** Which is the ground-state configuration of a selenium atom?

(A) [Ar] $4s^2$ $4p^4$
(B) [Kr] $4p^4$
(C) [Ar] $3d^{10}$ $4p^6$
(D) [Ar] $4s^2$ $3d^{10}$ $4p^4$

_____ **24.** How many unpaired electrons does a ground-state iron atom have?

(A) 4
(B) 0
(C) 2
(D) 6

_____ **25.** Which element has the ground-state configuration [Kr] $5s^2$ $4d^{10}$ $5p^3$?

(A) As
(B) Sb
(C) Bi
(D) In

_____ **26.** Which is the ground-state configuration of chromium?

(A) [Ar] $4s^2$ $3d^4$
(B) [Ar] $3d^6$
(C) [Ar] $4s^1$ $3d^5$
(D) [Ar] $4s^2$ $4p^4$

_____ **27.** How many valence electrons does a ground-state bromine atom have?

(A) 5
(B) 17
(C) 35
(D) 7

_____ **28.** Which configuration represents an atom in an excited state?

(A) $1s^2$ $2s^2$ $2p^6$
(B) $1s^2$ $2s^2$ $2p^5$ $3s^1$
(C) [Ne] $3s^2$ $3p^1$
(D) [He] $2s^1$

_____ **29.** Using Slater's rules, Zeff for the 3s electron of magnesium (Z = 12) is

(A) 2.00
(B) 9.15
(C) 3.20
(D) 2.85

_____ **30.** The valence electrons of which atom feel the largest effective nuclear charge?

(A) Na
(B) Al
(C) P
(D) Cl

_____ **31.** Which atom has the largest atomic radius?

(A) K
(B) Al
(C) Si
(D) Br

_____ **32.** Which order is correct for increasing radius of the isoelectronic ions Mg^{2+}, Na^{+}, F^{-} and O^{2-}?

(A) $O^{2-} < F^{-} < Na^{+} < Mg^{2+}$
(B) $Na^{+} < Mg^{2+} < O^{2-} < F^{-}$
(C) $Mg^{2+} < Na^{+} < F^{-} < O^{2-}$
(D) They are all the same size.

_____ **33.** Which best explains why Cl^{-} is larger than a Cl atom?

(A) Cl^{-} has more protons than Cl.
(B) Cl^{-} has an extra electron shell.
(C) The added electron increases electron–electron repulsion, so the cloud spreads out.
(D) Cl^{-} has a higher effective nuclear charge.

_____ **34.** Which is the ground-state configuration of Fe^{3+}?

(A) [Ar] $4s^2\ 3d^3$
(B) [Ar] $3d^5$
(C) [Ar] $4s^1\ 3d^4$
(D) [Ar] $3d^6$

_____ **35.** Which equation represents the second ionization energy of calcium?

(A) $Ca(g) \rightarrow Ca^{+}(g) + e^{-}$
(B) $Ca(g) \rightarrow Ca^{2+}(g) + 2e^{-}$
(C) $Ca^{+}(g) \rightarrow Ca^{2+}(g) + e^{-}$
(D) $Ca^{+}(g) + e^{-} \rightarrow Ca(g)$

_____ **36.** The successive ionization energies of element X are 496, 4562, 6910 and 9543 kJ/mol. X is most likely

(A) Mg
(B) Na
(C) Al
(D) Si

_____ **37.** Which atom has the highest first ionization energy?

(A) Na
(B) P
(C) K
(D) Ar

_____ **38.** Oxygen has a lower first ionization energy than nitrogen because

(A) oxygen has a smaller effective nuclear charge

(B) oxygen's fourth 2p electron is paired, and repulsion between the paired electrons makes it easier to remove

(C) oxygen is a larger atom than nitrogen

(D) nitrogen has a completely filled 2p subshell

_____ **39.** Which element has the most negative electron affinity?

(A) Na

(B) S

(C) Ar

(D) Cl

_____ **40.** Which pairing is correct?

(A) Highest first IE: F. Highest electronegativity: He.

(B) Highest first IE: He. Highest electronegativity: F.

(C) Highest first IE: Cs. Highest electronegativity: F.

(D) Highest first IE: He. Highest electronegativity: Cs.

Section II: Free Response

Directions: Show all work. Write the formula first, plug in values with units, and report answers with units and correct significant figures.

Question 1: Light and the hydrogen atom (10 points)

A sodium-vapor street lamp gives off yellow light with a wavelength of 589 nm.

(a) Calculate the frequency of this light. (2 points)

(b) Calculate the energy of one photon of this light. (2 points)

(c) Calculate the energy of one mole of these photons, in kJ/mol. (2 points)

(d) In a hydrogen atom, an electron drops from n = 7 to n = 2. Calculate ΔE for the electron and the wavelength of the light, in nm. (3 points)

(e) Is the light in part (d) absorbed or emitted? Justify your answer. (1 point)

Question 2: Magnesium and aluminum (10 points)

Answer the following about magnesium, aluminum and their ions. A periodic table may be used.

(a) Write the ground-state noble-gas configuration of Al. Then give one possible set of the four quantum numbers (n, l, m_l, m_s) for its highest-energy electron. (2 points)

(b) Using Slater's rules, calculate Zeff for the 3p electron of Al. (2 points)

(c) The first ionization energy of Al (578 kJ/mol) is lower than that of Mg (738 kJ/mol), even though Al has one more proton. Explain. (2 points)

(d) The first four ionization energies of Al are 578, 1817, 2745 and 11,577 kJ/mol. Explain the large jump between the third and fourth ionization energies. (2 points)

(e) Rank Na^{+}, Mg^{2+} and Al^{3+} from smallest to largest radius. Justify your ranking. (2 points)

Periodic Table of the Elements

1 **H** 1.01																	2 **He** 4.00
3 **Li** 6.94	4 **Be** 9.01											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.30											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.06	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.87	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.84	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.38	31 **Ga** 69.72	32 **Ge** 72.63	33 **As** 74.92	34 **Se** 78.97	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.95	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57 **La** 138.91	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.20	83 **Bi** 208.98	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 **Ac** (227)	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (266)	107 **Bh** (264)	108 **Hs** (277)	109 **Mt** (268)	110 **Ds** (281)	111 **Rg** (272)	112 **Cn** (285)	113 **Nh** (286)	114 **Fl** (289)	115 **Mc** (289)	116 **Lv** (293)	117 **Ts** (294)	118 **Og** (294)

*Lanthanides	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.36	63 **Eu** 151.96	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.04	71 **Lu** 174.97
†Actinides	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (262)

Atomic number above the symbol; average atomic mass below. Masses in parentheses are for the most stable isotope.

Answer Key: Practice Test A

Section I answers

1. C	**2.** A	**3.** D	**4.** B	**5.** C	**6.** A	**7.** D	**8.** A
9. C	**10.** A	**11.** C	**12.** D	**13.** B	**14.** B	**15.** A	**16.** D
17. C	**18.** A	**19.** C	**20.** B	**21.** D	**22.** C	**23.** D	**24.** A
25. B	**26.** C	**27.** D	**28.** B	**29.** D	**30.** D	**31.** A	**32.** C
33. C	**34.** B	**35.** C	**36.** B	**37.** D	**38.** B	**39.** D	**40.** B

Why each answer is right (and the tempting wrong one)

1. C Every kind of EM radiation moves at $c = 2.998\times10^{8}$ m/s in a vacuum. λ and ν are inversely related, so the first and last choices are backwards.

2. A Energy rises as wavelength shrinks: radio (longest λ) up to UV. The second choice is in order of decreasing energy.

3. D $\nu = c/\lambda = 3.00\times10^{8} / 6.00\times10^{-7} = 5.00\times10^{14}\ s^{-1}$. 2.00×10^{-15} is λ/c (the fraction flipped); 1.80×10^{2} is c × λ.

4. B $E = hc/\lambda$ with λ in meters: 400. nm = 4.00×10^{-7} m. The first choice forgot to convert nm to m.

5. C Halving λ doubles ν ($c = \lambda\nu$) and doubles E ($E = h\nu$). Energy and frequency always move together.

6. A Blue photons carry more energy than green ones, so they clear the minimum too. Red, infrared and microwave photons carry less.

7. D Brighter light means more photons, so more electrons, but each photon (and each electron's kinetic energy) is unchanged. Only a higher frequency raises the kinetic energy.

8. A Energy comes in quanta of hν. The uncertainty principle is about position and speed; Aufbau and Hund are filling rules.

9. C Excited gas atoms emit only certain wavelengths. Heated solids give continuous spectra.

10. A In a bond the electron clouds overlap, so half the internuclear distance is shorter. Two atoms that just touch keep their clouds separate.

11. C Longest λ means smallest energy. The levels crowd together as n grows, so the 4 → 3 drop is the smallest. 2 → 1 and 4 → 1 are the largest drops (shortest λ).

12. D Moving up (nf > ni) requires energy, so ΔE > 0 and a photon is absorbed.

13. B $-2.18\times10^{-18}\ J \div 2^{2} = -5.45\times10^{-19}\ J$. Dividing by 2 instead of 4 gives -1.09×10^{-18} J; the energy of a bound electron is negative.

14. B Bohr's model predicts hydrogen's lines but fails for atoms with more than one electron. Quantized levels and jumps between them are ideas that carry over to the modern model.

15. A That is why an orbital is a probability region, not a path. The second choice is the Pauli principle; the third is Aufbau.

16. D $l = 3 \le n - 1 = 3$ and $m_l = -3$ is within −l to +l. The others fail: $l = n$; $|m_l| > l$; m_s must be ±½.

17. C $2n^{2} = 2(16) = 32$. 16 is the number of orbitals (n^{2}).

18. A Any d subshell ($l = 2$) has $m_l = -2$ to +2: 5 orbitals. 10 is the number of electrons they hold.

19. C l (azimuthal) sets the shape: s, p, d, f. n sets size and energy; m_l sets orientation; m_s is spin.

20. B The number is n = 3; p means l = 1 (s = 0, p = 1, d = 2, f = 3).

21. D With more than one electron, 2s electrons spend time close to the nucleus and are shielded less, so 2s < 2p. 2s = 2p only in one-electron species like H and He^{+}.

22. C Hund's rule puts one electron in each equal-energy orbital before any pairing. Three parallel spins, all up or all down, is correct.

23. D Se has 34 electrons: 18 + 2 + 10 + 4. The first choice skips the $3d^{10}$ that fills before 4p.

24. A Fe = [Ar] $4s^2$ $3d^6$. Five 3d orbitals, six electrons: ↑↓ ↑ ↑ ↑ ↑, so 4 unpaired.

25. B 36 + 2 + 10 + 3 = 51: antimony, group 15 in period 5. As is the group-15 element in period 4.

26. C Cr is an exception: a half-filled $3d^5$ is extra stable, so one 4s electron moves into 3d. That gives 6 unpaired electrons.

27. D Br = [Ar] $4s^2$ $3d^{10}$ $4p^5$. The outermost shell (n = 4) holds $4s^2$ $4p^5$ = 7. The 3d electrons are not in the outer shell.

28. B A 2p electron has been promoted to 3s while 2p is still unfilled, so it breaks the Aufbau order. The others are ground states of Ne, Al and Li.

29. D S = 1(0.35) + 8(0.85) + 2(1.00) = 9.15, so Zeff = 12 − 9.15 = 2.85. 9.15 is S; 2.00 is the simple core-only method.

30. D Zeff on valence electrons rises across a period because protons are added while same-shell electrons shield poorly. Cl is farthest right.

31. A K is in period 4 (an extra shell compared with Al and Si) and at the far left of its period (low Zeff), so it is bigger than Br.

32. C All have 10 electrons. More protons pull harder: Mg (12) < Na (11) < F (9) < O (8). Same electron count does not mean same size.

33. C Same 17 protons, one more electron: more repulsion and less pull per electron. The extra electron goes into the same 3p subshell, not a new shell.

34. B Fe = [Ar] $4s^2$ $3d^6$. Remove both 4s electrons first, then one 3d: [Ar] $3d^5$. [Ar] $3d^6$ is Fe^{2+}.

35. C The 2nd IE removes one electron from the gaseous +1 ion. The first choice is the 1st IE; the second removes two electrons at once.

36. B The huge jump comes right after the 1st electron, so X has 1 valence electron: group 1, sodium.

37. D Ar is farthest right in period 3 (highest Zeff, smallest). K is lowest: bigger atom, valence electron farther out.

38. B N is $2p^3$ (half filled, all unpaired). O is $2p^4$; the paired electrons repel. O actually has the higher Zeff and is smaller.

39. D Cl is a halogen: one added electron completes 3p, releasing the most energy. Ar's EA is positive.

40. B He is the smallest atom with the most tightly held electrons, so it has the highest IE. F attracts bonding electrons most strongly, so it has the highest electronegativity. Cs is lowest in both.

Section II: Worked Answers

Question 1: Light and the hydrogen atom

(a) Calculate the frequency of this light. (2)

$\nu = c / \lambda$

$\lambda = 589\ \text{nm} \times (1\ \text{m} / 10^{9}\ \text{nm}) = 5.89\times10^{-7}\ \text{m}$

$\nu = (2.998\times10^{8}\ \text{m/s}) / (5.89\times10^{-7}\ \text{m}) = 5.09\times10^{14}\ \text{s}^{-1}$

1 point: setup with the formula and λ in meters. 1 point: answer with units and 3 sig figs.

(b) Calculate the energy of one photon of this light. (2)

$E = h\nu$

$E = (6.626\times10^{-34}\ \text{J·s})(5.09\times10^{14}\ \text{s}^{-1}) = 3.37\times10^{-19}\ \text{J}$

($E = hc/\lambda$ gives the same result.)

(c) Calculate the energy of one mole of these photons, in kJ/mol. (2)

$E(\text{mol}) = E(\text{photon}) \times N_a$

$E = (3.37\times10^{-19}\ \text{J})(6.022\times10^{23}\ \text{mol}^{-1}) = 2.03\times10^{5}\ \text{J/mol}$

$2.03\times10^{5}\ \text{J/mol} \times (1\ \text{kJ} / 1000\ \text{J}) = 203\ \text{kJ/mol}$

(d) In a hydrogen atom, an electron drops from n = 7 to n = 2. Calculate ΔE for the electron and the wavelength of the light, in nm. (3)

$\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/n_f^2 - 1/n_i^2)$

$\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/2^2 - 1/7^2) = -2.18\times10^{-18}\ \text{J}\ (0.2296) = -5.01\times10^{-19}\ \text{J}$

$\lambda = hc / |\Delta E| = (6.626\times10^{-34}\ \text{J·s})(2.998\times10^{8}\ \text{m/s}) / (5.01\times10^{-19}\ \text{J}) = 3.97\times10^{-7}\ \text{m}$

$3.97\times10^{-7}\ \text{m} \times 10^{9}\ \text{nm/m} = 397\ \text{nm}$

(e) Is the light in part (d) absorbed or emitted? Justify your answer. (1)

Emitted. ΔE is negative because the electron drops to a lower energy level ($n_f < n_i$), and the energy it loses leaves as a photon.

Question 2: Magnesium and aluminum

(a) Write the ground-state noble-gas configuration of Al. Then give one possible set of the four quantum numbers (n, l, m_l, m_s) for its highest-energy electron. (2)

Al: [Ne] $3s^2\ 3p^1$

Highest-energy electron is 3p: $n = 3$, $l = 1$, $m_l = -1$, 0 or +1 (any one), $m_s = +½$ (or $-½$)

(b) Using Slater's rules, calculate Zeff for the 3p electron of Al. (2)

$Z_{eff} = Z - S$

Al = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$. Other n = 3 electrons: 2 (the $3s^2$); n = 2: 8; n = 1: 2

$S = 2(0.35) + 8(0.85) + 2(1.00) = 0.70 + 6.80 + 2.00 = 9.50$

$Z_{eff} = 13 - 9.50 = 3.50$

(c) The first ionization energy of Al (578 kJ/mol) is lower than that of Mg (738 kJ/mol), even though Al has one more proton. Explain. (2)

Al's first electron comes out of a 3p orbital; Mg's comes out of 3s.

The 3p electron is higher in energy and is shielded by the $3s^2$ electrons, so it is held less tightly and takes less energy to remove.

(d) The first four ionization energies of Al are 578, 1817, 2745 and 11,577 kJ/mol. Explain the large jump between the third and fourth ionization energies. (2)

After three electrons are removed, Al^{3+} is [Ne]: all three valence electrons ($3s^2\ 3p^1$) are gone.

The fourth electron comes from the n = 2 core, which is much closer to the nucleus and feels a much larger Zeff, so far more energy is needed.

(e) Rank Na^+, Mg^{2+} and Al^{3+} from smallest to largest radius. Justify your ranking. (2)

$Al^{3+} < Mg^{2+} < Na^+$

All three are isoelectronic (10 electrons, [Ne]). Al has the most protons (13, then Mg 12, then Na 11), so its electrons feel the highest Zeff and are pulled closest to the nucleus.